



2026 BASIC ABSTRACT ALGEBRA STUDY

LECTURE 1 HOMEWORKS

QUESTION PAPER

Version

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Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$ for an integer n . i.e., the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B , A is said to be a subset of B if the following condition holds.

If x is an element of A , then x is an element of B .

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A . This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.

EASY

Which of the following is distinct from the others?

Let A and B be two sets. Let $A(x)$ mean x is an element of A , and $B(x)$ mean x is an element of B .

Which of the following is equivalent to saying that $A \subset B$?

- ☐ 1. $B(x) \rightarrow A(x)$
☐ 2. $\text{not } A(x) \rightarrow B(x)$
☐ 3. $\text{not } A(x) \rightarrow \text{not } B(x)$
- ☐ 4. $A(x) \rightarrow \text{not } B(x)$
☐ 5. $\text{not } B(x) \rightarrow \text{not } A(x)$

Q2.

EASY

Let Ψ_2 denote the set of all complex solutions to the equation $x^2 = 1$ and let Ψ_4 denote the set of all complex solutions to the equation $x^4 = 1$.

Show that $\Psi_2 \subset \Psi_4$.

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m .

Hint 1. Provide concrete n and m that make up a counterexample. Try justifying your answer using the definition of set builder notation.

Q4.

HARD

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 1. cf. lecture note.

Q5.

HARD

For a positive integer n , let Ψ_n denote the set of all complex solutions to the equation $x^n = 1$.

Show by using Definition 2 that for any two positive integers n and m , $\Psi_n \subset \Psi_{nm}$.

Hint 1. Of course, it is a generalization of Q2.

Q6.

HARD

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n .

Hint 1. cf. PCSA Q4.

Q7.

HARD

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n , then either $n = 1$ or $n = -1$.

Hint 1. What are the factors of 1 and -1 ?

Definition 2 (Symmetric difference)

$A \triangle B$ is a binary set operation given by $(A \setminus B) \cup (B \setminus A)$ and is called a *symmetric difference* between A and B .

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that $A \triangle B \subset C$ if and only if $A \cap B \subset (A \cap B) \cup C$.

LEVI

Q9.

For an integer n , let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

- (a) Show that every two distinct elements of P are disjoint.

Hint 1. Use Definition 3 and contraposition.

- (b) Show that the following holds for any two integers a and b by Definition 3.

$$M(a) + M(b) = M(a + b)$$

Q10.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2^{121}} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: _____